Anticipated Defence Questions and Answers

Measuring and Prioritising Technical Debt in Mission-Critical Trading Systems

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Sections

Prepared for the defence of "Measuring and Prioritising Technical Debt in Mission-Critical Trading Systems". Every answer is grounded in the thesis; chapter references are given so each claim can be located under pressure. Five of the questions (small-n, circularity, proxy validity, weight choice and team misuse) are also addressed on slides 13 and 14; the remainder are held in reserve.

Opening and framing

Q1. Summarise your thesis in a few sentences. Vendor-dominated trading IT carries significant technical debt outside source code, where the dominant code-centric tools cannot measure it, yet its symptoms are recorded daily in ITSM and CMDB systems. This thesis designs and evaluates a TOPSIS-based framework that converts five such operational indicators into a service-level debt priority ranking, demonstrated on two years of live records from a regulated Swiss trading institution. A five-strand evaluation shows the ranking is robust at the extremes across the entire weight space, adds information over an incident count, and is largely method-independent, while the one systematic method disagreement itself yields a transferable finding about how ranking methods treat concentrated debt profiles.

Q2. Why does this problem matter? Who benefits? Because the debt in question produces measurable harm and current governance cannot see it. Banker and colleagues quantify the firm-level cost, with ten per cent more debt costing sixteen per cent of gross return on assets in COTS platforms; Di Tizio and colleagues show one month of patch delay multiplying compromise odds by 4.9. In a trading institution these consequences meet continuous availability obligations and FINMA oversight, yet remediation decisions are made largely on judgement and incident counts, which the discriminant analysis shows miss exactly the structural exposure that matters, as SVC-02 demonstrates. The beneficiaries are IT portfolio governance in regulated institutions, who gain an auditable, reproducible priority signal from data they already hold (Chapters 1, 2 and 6).

Q3. What is your original contribution to knowledge? Three contributions, stated in Chapter 6. First, to the best of current knowledge, the first application of TOPSIS to live ITSM and CMDB records for technical debt prioritisation in financial trading IT, closing a combination gap the literature review documents: service-level indicators, multi-criteria prioritisation, live operational data and a trading context each exist separately, never together. Second, an operationally defined indicator set with an explicit literature mapping to debt dimensions, together with evidence of its behaviour under a multi-strand evaluation. Third, a methodological finding of independent interest: sample-relative and absolutely anchored MCDM methods respond differently to concentrated debt profiles, and the difference is consequential at the top of a ranking.

Q4. How do you define technical debt, and what is in and out of scope? The thesis starts from Cunningham's metaphor and the subsequent taxonomies that broadened debt beyond source code, and it deliberately targets the non-code dimensions that dominate vendor-managed environments: architectural, infrastructure/platform, process, vulnerability and infrastructure lifecycle debt, as mapped in Table 2.3. Code-level debt is out of measurement scope, not by judgement of importance but by access, since source code is unavailable, and the mapping table explicitly declares the dimension assignments as this study's synthesis. Equally important is what the framework does not claim: it does not measure debt as a quantity; it orders services by debt's operational consequences, an ordinal signal, as stated from Chapter 1 onward.

Q5. Why is the service the unit of analysis, rather than the application, component or team? Three reasons from Chapter 3. Decisions are made at that level: remediation budgets, vendor negotiations and risk acceptance in the case organisation attach to services as configuration items. Data exists at that level: ITSM incidents and changes and CMDB lifecycle records are keyed to the service CI, whereas component-level detail inside vendor platforms is precisely what the organisation cannot see. And governance requires it: the agreed scope describes services, not teams, which the ethics position depends on, since high-debt profiles typically reflect historical design decisions rather than current custodians.

Method and design

Q6. Why Design Science Research, and not a survey, an experiment or an interpretive case study? Because the research output is an artefact, not a theory. The thesis does not primarily test a hypothesis about the world or interpret practitioners' experiences; it constructs, demonstrates and evaluates a working prioritisation instrument, and DSR is the paradigm whose success criterion is the demonstrated utility of a designed artefact [Hevner et al., 2004]. The alternatives fail on access or on output. A survey or experiment cannot reach the granular, longitudinal ITSM and CMDB records the indicators require, and would yield perceptions rather than operational evidence. A purely interpretive case study could characterise the debt but would produce no instrument, leaving the gap identified in Chapter 2 open. Action research would have required intervening in live trading operations across repeated cycles, which sits outside the governance scope agreed with the organisation (Chapter 3). DSR also fits the pragmatist stance the methodology declares: the framework is judged by whether it produces a defensible, reproducible ordering an institution can act on, and the seven Hevner guidelines supplied the evaluation discipline, including the analytical evaluation strategy of Guideline 3, that structures Chapter 5. In one breath: the question demanded a built and evaluated artefact from privileged operational data, and DSR is the methodology designed for exactly that output.

Q7. Why TOPSIS rather than AHP, VIKOR or a fuzzy extension? Because the task is producing a full ranked list of a service portfolio from mixed cost and benefit criteria on different scales, and TOPSIS integrates normalisation, weighting and ranking in one transparent procedure. AHP derives weights from pairwise comparisons but does not natively produce a full ranking, and its comparison burden grows with the number of criteria. VIKOR identifies compromise solutions rather than a global ranking. Crucially, the thesis does not treat this choice as settled by argument: method choice is handled as a robustness question, and the evaluation re-ranks everything with SAW and VIKOR precisely to test whether the ordering depends on the mechanics of one method (Chapters 2 and 5). Fuzzy extensions address imprecise expert judgements; the inputs here are measured operational values, so the added machinery would not address a problem this study has.

Q8. Why vector normalisation rather than min-max? Min-max rescales to fixed bounds and compresses extreme observations, altering the relative distances on which a distance-based method relies. Vector normalisation preserves proportional relationships, so a service with an unusually high MTTR keeps its full separation in the normalised space. Greco and colleagues recommend it for composite indicators for exactly this reason (Chapter 4). The one place min-max is used deliberately is SAW in the convergent-validity strand, where Weitendorf normalisation avoids the division by zero that classic cost normalisation would produce on the dataset's genuine zero values (Chapter 5).

Q9. Your differential weights are not derived from literature. Are they arbitrary? They are a declared practitioner assumption, stated as such in Chapter 3: recovery time and patching delay carry greater operational consequence in time-critical trading environments. The design response to that arbitrariness is twofold. The two discrete configurations bound the effect, and the Monte Carlo strand removes the dependence entirely: across 20,000 weight vectors drawn uniformly from the simplex, SVC-03 is last in 98.1 per cent of weightings, and rank 1 never leaves the SVC-01, SVC-04, SVC-06 cluster. The conclusions the thesis actually relies on hold under any admissible weighting.

Q10. Why a uniform Dirichlet for the weight sweep? Because absent evidence privileging one weighting over another, every valid weight vector should be equally plausible; a uniform draw over the simplex encodes exactly that neutrality. The procedure approximates, in tractable form, the stochastic acceptability analysis Greco and colleagues advocate for composite indicators. The run is seeded and reproducible, and repeating it with a different seed and sample size changed no reported figure by more than 0.01 (Chapter 5).

Q11. Why Kendall's tau rather than Spearman's rho? Tau has a direct decision-relevant interpretation for this problem: it counts pairwise order agreements, and a pairwise inversion is precisely what changes a remediation decision. With six alternatives it also admits an exact p-value by enumerating all 720 orderings, avoiding any large-sample approximation the data could not support. The p-values are read as effect sizes rather than accept-or-reject decisions, because the sample size caps attainable significance regardless of how strong the agreement is (Chapter 5).

Data

Q12. Six services from a portfolio of 108. How were they chosen, and what can that support? Selection was by data completeness: each of the six holds a continuous record across all five indicators for the full two-year window in both the ITSM instance and the CMDB, is an active configuration item, and has operated for more than thirty-six months. Fifty-three small vendor interfaces requiring no significant ongoing work were excluded first. Two consequences are declared in Chapter 4: the selected services skew toward higher operational activity, so absolute values are not portfolio-representative, and six alternatives limit the discriminatory power of the rank-stability tests. The contribution is therefore the instrument and its evaluation logic, not a portfolio ranking; portfolio-wide deployment is stated future work.

Q13. Your change failure rate takes values above one. That is not a rate. Explain. Correct, and Chapter 3 declares it. CFR here is a derived proxy, weighted incidents divided by the number of changes, which is the formula the case organisation itself applies in practice. It approximates the intent of the DORA change failure rate without being identical to it, because a true CFR requires deployment-outcome labelling the organisation does not record. The values function correctly as a cost criterion in the model; only the label inherits DORA vocabulary.

Q14. CFR shares its numerator with incident frequency. Is that not double counting? The two criteria are not fully independent, and the thesis says so. The overlap is accepted as a constraint of working with operational data rather than engineered away, and its effect on the ordering is bounded by the weight sensitivity analysis: rankings are stable when the weight on either criterion moves, which they could not be if the shared numerator were silently driving the result (Chapters 3 and 5).

Q15. Why cap MTTR at the 98th percentile? Because the upper tail of the resolution-time distribution is dominated by records whose elapsed time reflects administrative inactivity rather than active resolution effort, for example tickets left open across calendar time. Including them would let data-hygiene artefacts, not debt, drive a criterion. The cap is a declared data-preparation decision in Chapter 4, alongside alarm-storm consolidation and removal of automatic closures under 0.1 hours.

Q16. Why a two-year observation window, January 2024 to December 2025? It is the longest span over which all five indicators hold continuous, complete records for the demonstration services in both the ITSM instance and the CMDB, and completeness was the selection criterion throughout. Two full years also enable the temporal strand of the evaluation, the 2024 versus 2025 comparison, which a single-year snapshot could not support, and annual windows align with the organisation’s own reporting cadence. A longer horizon is desirable and is exactly what the longitudinal future-work direction describes; it was not available at the required data quality for this study (Chapters 3, 4 and 6).

Validity

Q17. You built and evaluated the framework on the same dataset. Is the evaluation circular? The design is declared and its consequences are bounded rather than hidden. What the evaluation establishes is robustness, method-independence and discriminant value; what it cannot establish is criterion validity, because no independent measure of the non-code debt in scope exists to validate against, a gap that is itself documented in the literature review. Proxy validity is therefore assessed on convergence with theory rather than demonstrated causal mechanism, and external validation in other organisations is named as the primary avenue for future work (Chapters 3, 5 and 6).

Q18. Degraded operational metrics could reflect staffing, release practices or organisational disruption rather than debt. How do you exclude that? The thesis does not claim to exclude it; it names those exact confounders in Chapter 2 and treats the proxy link as the central testable hypothesis rather than a premise. Three things make the hypothesis credible within this design: Nord and colleagues ground elevated MTTR and CFR as diagnostic debt symptoms rather than generic shortfalls; the multi-criteria construction means no single confounded metric drives a rank; and the resulting profiles match what the literature predicts, for instance SVC-01’s incident-quiet, lifecycle-heavy profile. A cross-sectional design cannot demonstrate the causal mechanism, which is why longitudinal tracking is the named next step.

Q19. What would falsify the proxy hypothesis? Longitudinal disconfirmation: if services ranked high priority systematically failed to generate the operational and financial consequences the debt literature predicts, in the sense of Banker and colleagues, while low-priority services generated them, the proxy claim would fail. Chapter 6 frames longitudinal evaluation in exactly these terms, as the study that would move the claim from plausible to evidenced or refute it.

Q20. Two of your key sources are single-organisation studies. Does your evidence base generalise? The thesis flags this itself: the Credit Suisse, BankAlpha and ING studies are each single-organisation, and whether their patterns generalise to securities trading remains open. This study does not resolve that; it contributes an operational dataset from a trading environment that was previously absent from the literature, which is a necessary step toward the multi-site evidence base the field lacks (Chapter 2).

Q21. This is a single-organisation study. What can it tell anyone else? The claim is calibrated to the design. What transfers is the artefact and its evaluation logic: the indicator set with its literature mapping, the construction pipeline, and the five-strand evaluation template, all of which are specified independently of this organisation. What does not transfer without revalidation is the calibration: the weights, the absolute-mode bounds and the recording-practice assumptions are local, and Chapter 6 says so. The case was chosen because its characteristics, vendor-dominated architecture, continuous availability and FINMA-class regulation, define a recognisable class of institutions, so external validation across that class is the named first direction for future work rather than an afterthought (Chapters 3 and 6).

Results

Q22. What actually happened to SVC-06 between 2024 and 2025? It deteriorated broadly, and the framework tracked it. In 2024 SVC-06 sat fifth under every method: 43 incidents, the second-lowest count, a change failure rate of 0.86, and a below-par but not worst patch recency of 82.35. By 2025 its incident count had risen to 302, its change failure rate had become the portfolio's worst at 3.96, and its patch recency had fallen to 52.94, the lowest but for SVC-01. It rose from fifth to third under equal weights, and every method places it in the 2025 top three. SVC-06 is therefore the temporal-validity case: a genuine, broad deterioration that the ranking followed, which is exactly the behaviour a faithful instrument should show and why cross-year movement is reported as descriptive context rather than penalised as instability (Chapter 5, Appendices B and C).

Q23. So which service actually gets remediated first, SVC-04 or SVC-06? For 2025 the answer is unusually clear: every method under equal weights puts SVC-01 first, so remediation starts with its infrastructure lifecycle exposure. The contest is visible elsewhere. In 2024 under equal weights all methods agree that SVC-04 and SVC-06 are the two highest-priority services and split only on which leads, TOPSIS naming SVC-04 and SAW and VIKOR naming SVC-06, a split decided by normalisation rather than by any real gap. The Monte Carlo analysis reinforces this, with rank 1 in 2025 never leaving the SVC-01, SVC-04, SVC-06 cluster but splitting within it (46, 27 and 27 per cent). Whether concentrated structural exposure outranks diffuse operational debt is a compensability judgement about the organisation's risk appetite, informed by the dimensional profiles the framework supplies. The thesis regards declining to fabricate a numeric tiebreak as intellectual honesty, not indecision (Chapters 5 and 6).

Q24. Is rank reversal actually a problem at n = 6, or a theoretical worry? It is observable and consequential in this data: the order changes in two of six removals under the standard formulation, and the changes reach the very top, since removing SVC-04 lifts SVC-06 above SVC-01 into first place and removing SVC-06 swaps the top two. The lowest rank never moves. The absolute-mode correction eliminates the effect entirely across all removals in both years, and not incidentally: Yang proves that variants with set-independent bounds and ideal solutions are ranking stable by construction. The cost is that the fixed bounds must be justified from organisational thresholds, which is why the variant is a complementary perspective rather than the primary signal (Chapter 5, Appendix A).

Q25. Portfolios churn: services retire and new ones arrive. Does that not make absolute TOPSIS the necessary primary method? No, for three reasons, though churn does make the complementary role more important. First, re-ranking after churn is correct behaviour, not rank reversal: prioritisation is a competition, and a competition with different entrants legitimately has a different result; the pathology is unmotivated flips among unchanged services, and the leave-one-out test shows this risk is real in the data: the classic top rank itself flips in two of six removals, which is exactly why the absolute variant exists to certify the primary ranking. Second, absolute mode is only churn-proof while its fixed bounds hold; a new arrival outside the calibrated bounds forces either clipping, which destroys discrimination where the signal is strongest, or bound revision, which shifts every score at once and readmits instability as a governance problem. Third, churn makes spiky new profiles more likely, which is exactly the shape the compensatory absolute variant masks, as SVC-06 in 2024 shows. In deployment the division of labour therefore holds: sample-relative methods answer who is first in today's portfolio, while the absolute frame tracks whether an individual service is deteriorating on a scale that does not move (Chapter 5).

Q26. If you had full source code access, would this framework still be needed? Yes, because the two views measure different debt. Code analysis quantifies code quality debt; it does not see unsupported component exposure, patch delay, or the operational symptoms of architectural and process debt, which are the dimensions this framework covers and which the literature identifies as understudied precisely because tooling is code-bound. The converse also holds, which is why the thesis claims complementarity rather than replacement: with code access, the natural extension is adding code-derived indicators, such as the excluded test coverage and coupling metrics, as further criteria in the same multi-criteria structure, not abandoning the operational view. Access would remove a constraint on the indicator set; it would not remove the need for a service-level, portfolio-wide, auditable prioritisation instrument (Chapters 2 and 3).

Governance, ethics and impact

Q27. Could this instrument be used to evaluate teams, and what prevents that? Three safeguards. The governance scope agreed with the Head of Trading IT at the outset defines the output as decision support for portfolio conversations, not a binding assessment instrument. The dataset contains no team, vendor or individual identifiers, with anonymisation applied before analysis. And the thesis states the interpretive principle: high-debt profiles typically reflect historical design decisions rather than the competence of current custodians, following Ahmad and colleagues. The framework describes services, not people (Chapters 3 and 6).

Q28. What does FINMA Circular 2023/1 have to do with a technical debt ranking? The circular requires comprehensive ICT operational risk management, including incident monitoring and service resilience, and it charges the board with approving and monitoring the relevant strategies. It does not prescribe the metrics used here; those are organisationally defined. The connection is that an auditable, reproducible priority ordering built from the institution's own operational records is precisely the kind of demonstrable risk-management evidence that regime rewards, which is why auditability is a design requirement rather than a nicety (Chapters 3 and 6).

Q29. What would an organisation need to adopt this framework? Three things, all stated in the thesis. An ITSM instance and CMDB with sufficient recording discipline to populate the five indicators, which is the binding constraint, since indicator coverage limited even this study to six services. A locally justified weight configuration, revalidated rather than transferred, because the single-site calibration is a declared limitation. And a governance agreement on use, of the kind established here, so the ordinal signal feeds portfolio conversations rather than performance management (Chapters 3, 4 and 6).

Q30. Where does this work go next, academically? Four named directions in Chapter 6: external validation in other regulated organisations to test the proxy hypothesis beyond one case; longitudinal evaluation to test the predictive content of the rankings; portfolio-wide deployment once indicator coverage is complete, which also strengthens the statistical footing; and deriving the absolute-mode bounds from regulatory or cross-industry reference points, which would make scores comparable between organisations as well as between years. For publication, the methodological finding, that sample-relative and absolutely anchored methods respond differently to concentrated profiles, stands on its own for an MCDM venue, while the framework and case evidence suit an empirical software engineering or IS venue.

Reflection

Q31. What is the weakest part of the thesis? The absence of criterion validity, and the thesis says so itself rather than waiting to be told. Every evaluation strand draws on the same dataset used to construct the framework, and no independent measure of non-code debt exists to validate against, so the evaluation establishes robustness, method-independence and discriminant value and explicitly stops there. The second candidate is the six-service demonstration set, which caps the statistical power of the rank tests. Both weaknesses are declared, bounded and converted into the two leading future-work directions, external and longitudinal validation, which is the strongest available response to a limitation that the state of the field, not the study design, imposes (Chapters 5 and 6).

Q32. With hindsight, what would you do differently? Two things, both about data groundwork rather than method. I would have engaged the organisation earlier about deployment-outcome labelling, which would have permitted a true DORA change failure rate instead of the derived proxy and removed the shared-numerator dependency with incident frequency. And I would have pushed sooner on indicator coverage across the wider portfolio, since completeness, not method, is what limited the demonstration to six services. I would not change the method core: the decision to treat method choice itself as a robustness question is what produced the SVC-06 finding, which turned out to be one of the most valuable results.

Q33. What surprised you during the research? Two things, recorded in the reflective note. The main difficulty was definitional rather than computational: deciding what an incident count or a derived change failure rate can legitimately mean consumed more effort than implementing the model. And the alternative methods, expected to be a confirmation exercise, disagreed about SVC-06 in a way that forced an explicit position on whether debt dimensions are compensatory, converting a validation step into a finding (Chapter 6).